



Southeast University

Introduction to Software Engineering

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Outline

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Part 2. Modeling

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- ❖ **2. Structured Analysis**
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 - **The definition of classes**
 - **CRC card**
 - **Analysis Packages**

Requirements Modeling Strategies

- ❖ One view of requirements modeling, called *structured analysis*, considers data and the processes that transform the data as separate entities.
 - Data objects are modeled in a way that defines their attributes and relationships.
 - Processes that manipulate data objects are modeled in a manner that shows how they transform data as data objects flow through the system.
- ❖ A second approach to analysis modeled, called *object-oriented analysis*, focuses on
 - the definition of classes and
 - the manner in which they collaborate with one another to effect customer requirements.

需求建模策略

Requirements Modeling Strategies

- 一种需求建模视图，称为结构化分析，认为数据和转换数据的过程是独立的实体。
 - 数据对象以定义其属性和关系的方式进行建模。
 - 操作数据对象的过程以显示它们如何在系统中作为数据对象流动的方式进行建模。
- 第二种分析建模方法，称为面向对象分析，关注
 - 类的定义以及
 - 它们如何相互协作以影响客户需求。

○ 基于类建模 ☆ ☆

1. 结构化分析——将数据和过程转换成独立的实体（具有分层结构） ☆

- 数据对象用属性+关系来描述
- 过程用在系统中流转数据的方式来描述

Structured Analysis

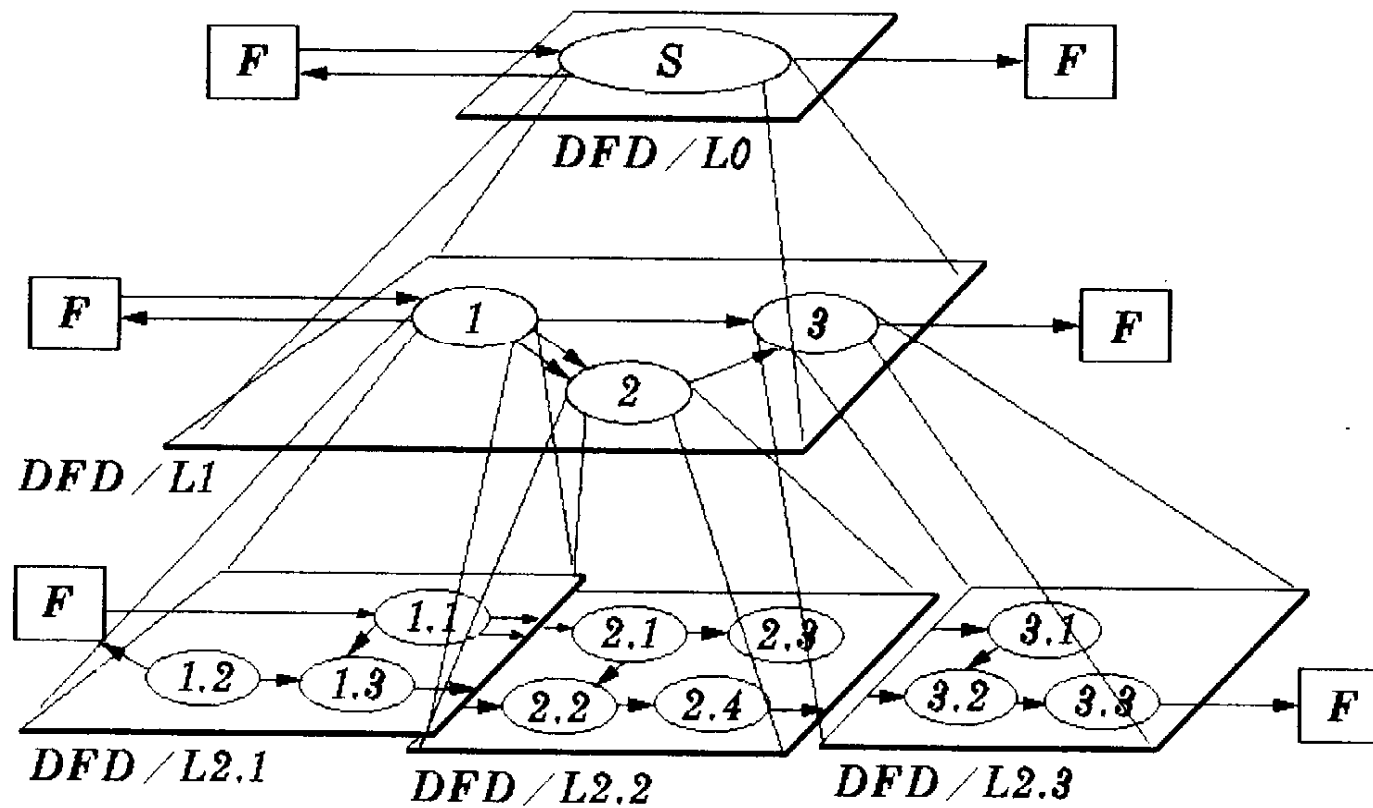
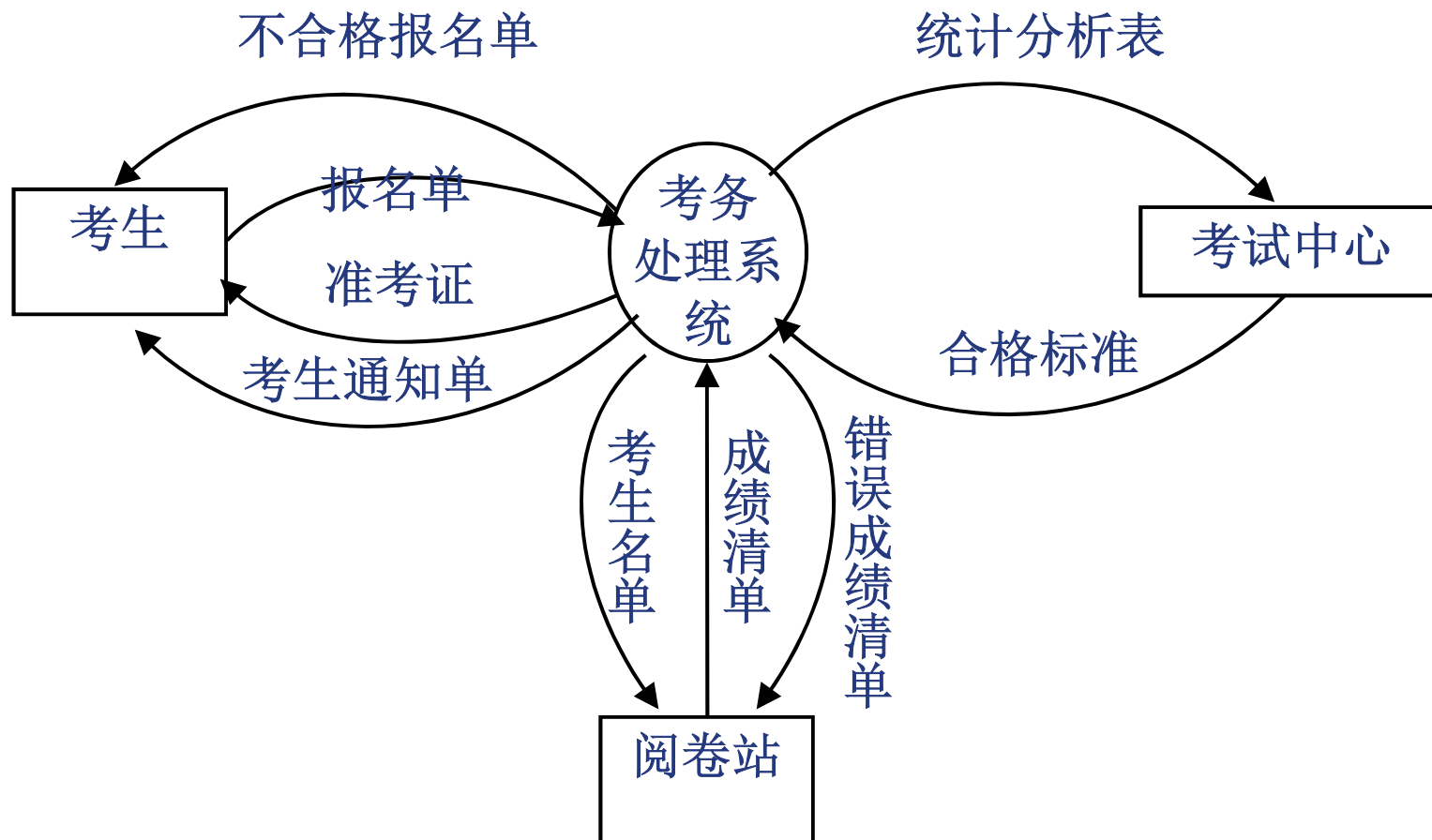


图 2.9 分层数据流图

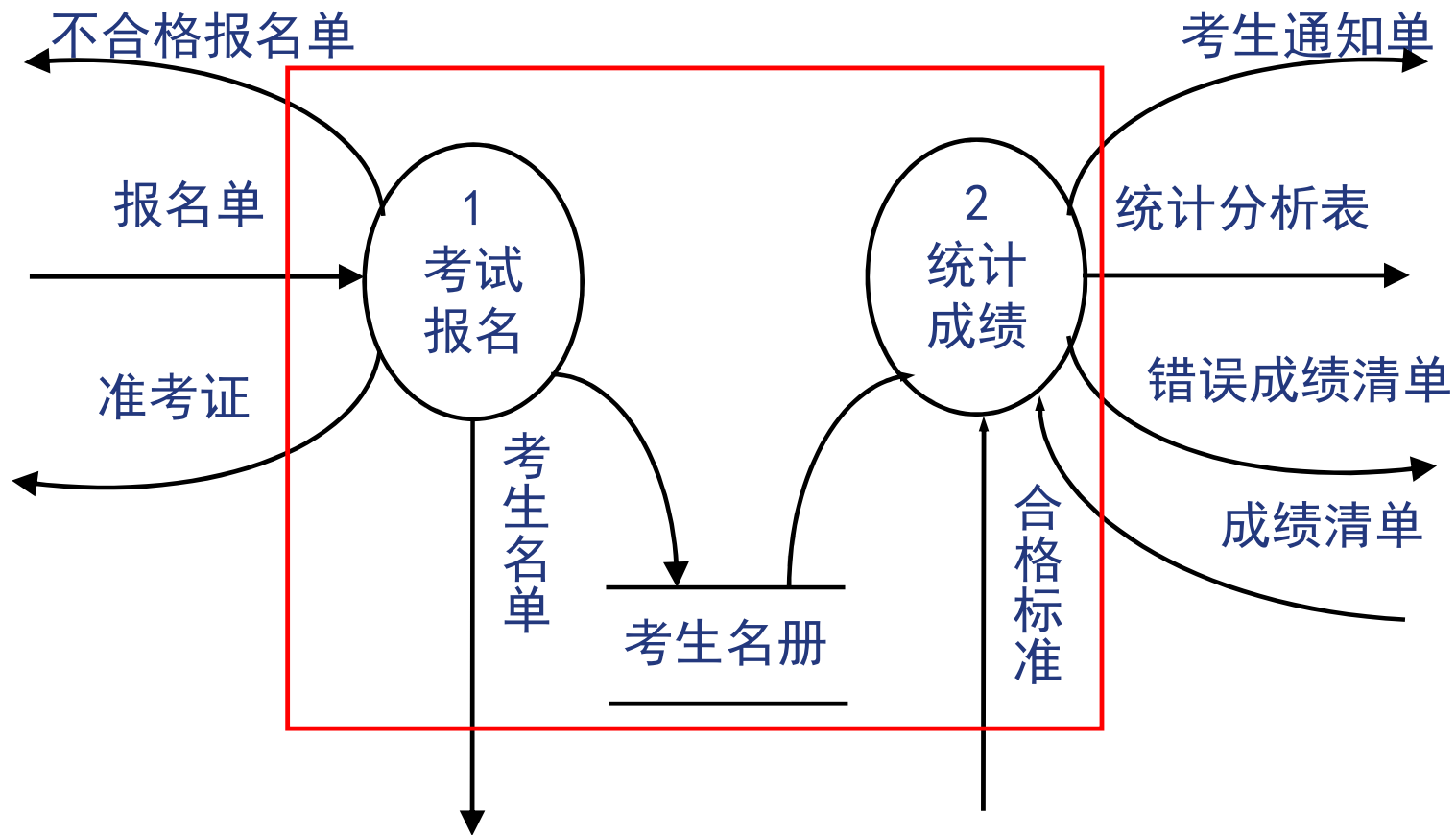
Structured Analysis

❖ Example: 考务处理系统顶层图



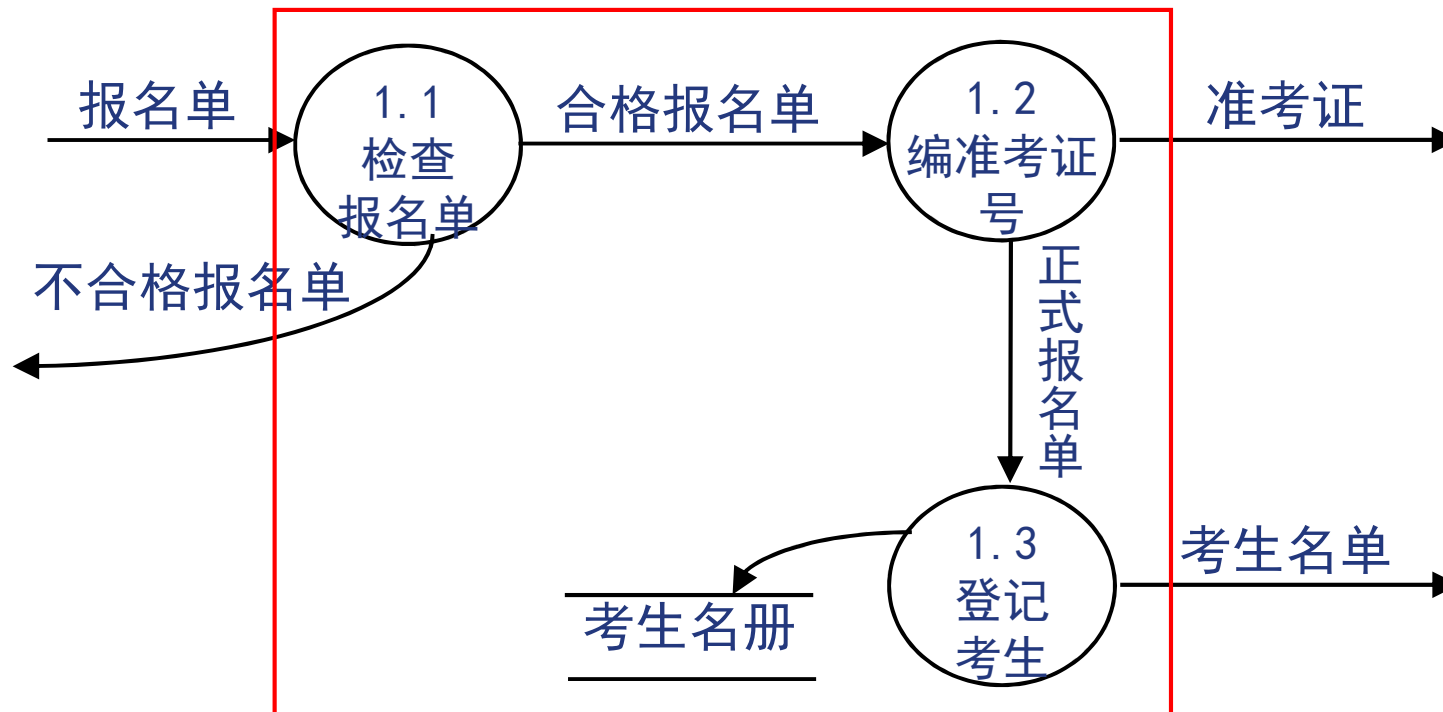
Structured Analysis

考务处理系统1层图



Structured Analysis

考务处理系统2层图：考试报名



Class-Based Modeling

❖ Class-based modeling represents:

- **objects** that the system will manipulate
- **operations** (also called methods or services) that will be applied to the objects to effect the manipulation
- **relationships** (some hierarchical) between the objects
- **collaborations** that occur between the classes that are defined.

❖ The elements of a class-based model include **classes and objects, attributes, operations, CRC models, collaboration diagrams and packages.**

- 基于类的建模表示：
 - 基于类的模型的要素包括类和对象、属性、操作、CRC模型、协作图和包。
 - 系统将操作的对象
 - 将应用于对象以进行操作的操作（也称为方法或服务）
 - 对象之间的关系（有些是层次结构的）
 - 在定义类之间发生的协作

识别分析类

- 提取名词
 - 潜在类的判别：☆
 1. 保留信息 ——记录该类的信息才能保证系统正常工作
 2. 需要操作 ——对该类具有一组操作用以改变类的属性值
 3. 多个属性 ——只有一个属性的类最好作为另一个类的某个属性
 4. 公共属性 ——该类具有一组属性适用于所有实例
 5. 公共操作 ——该类具有一组操作适用于所有实例
 6. 必须要求——问题空间外的实体或者解决问题中产生的信息
 - 潜在类的分类
 1. 外部实体——产生或使用系统信息
 2. 事物——信息的一部分
 3. 事件——系统中发生的操作
 4. 角色——和系统交互的人员
 5. 组织单元——与某个应用有关
 6. 场地——问题的环境和功能
 7. 结构——对象的类或相关的类
- 画出类图——简单->属性、操作->约束
 - >操作包含：增删改、计算、查、监视

Identifying Analysis Classes

- ❖ Examining the usage scenarios developed as part of the requirements model and perform a "grammatical parse" [Abb83]
 - Classes are determined by underlining **each noun or noun phrase** and entering it into a simple table.
 - Synonyms should be noted.
 - If the class (noun) is required to implement a solution, then it is part of the solution space; otherwise, if a class is necessary only to describe a solution, it is part of the problem space.

- ❖ But what should we look for once all of the nouns have been isolated?

- 检查作为需求模型一部分开发的使用场景，并执行“语法解析” [Abb83]
 - 通过下划线标注每个名词或名词短语，并将其输入到一个简单的表格中来确定类。
 - 应注意同义词。
 - 如果类（名词）是实现解决方案所需的，那么它是解决方案空间的一部分；否则，如果类仅用于描述解决方案，则它是问题空间的一部分。
- 但是，一旦所有名词都被隔离出来，我们应该寻找什么呢？

Manifestations of Analysis Classes

❖ *Analysis classes* manifest themselves in one of the following ways:

- *External entities* (e.g., other systems, devices, people) that produce or consume information
- *Things* (e.g., reports, displays, letters, signals) that are part of the information domain for the problem
- *Occurrences or events* (e.g., a property transfer or the completion of a series of robot movements) that occur within the context of system operation
- *Roles* (e.g., manager, engineer, salesperson) played by people who interact with the system
- *Organizational units* (e.g., division, group, team) that are relevant to an application
- *Places* (e.g., manufacturing floor or loading dock) that establish the context of the problem and the overall function
- *Structures* (e.g., sensors, four-wheeled vehicles, or computers) that define a class of objects or related classes of objects

分析类的表现形式

Manifestations of Analysis Classes

- 分析类以以下方式之一表现出来：
 - 外部实体（例如，其他系统、设备、人员）生产或消耗信息
 - 事物（例如，报告、显示、信件、信号）是问题信息域的一部分
 - 发生或事件（例如，财产转移或一系列机器人动作的完成）在系统操作上下文中发生
 - 角色（例如，经理、工程师、销售人员）由与系统交互的人扮演
 - 组织单位（例如，部门、团队、组）与应用相关
 - 地点（例如，制造车间或装卸区）建立问题上下文和整体功能
 - 结构（例如，传感器、四轮车或计算机）定义对象类或相关对象类

- ❖ **Safehome software** enables **the homeowner** to configure **the security system** when it is installed, monitors all **sensors** connected to the security system, and interacts with the homeowner through **the Internet** or **a control panel**.
- ❖ During **installation**, the SafeHome control panel is used to “program” and configure the system. Each sensor is assigned **a number** and **type**, **a master password** is programmed for arming and disarming the system, and **telephone number(s)** are input for dialing when **a sensor event** occurs.
- ❖ When a sensor event is sensed by the software, it rings **an audible alarm** attached to the system. After **a delay time** that is specified by the homeowner during **system configuration activities**, the software dials a telephone number of **a monitoring service**, provides information about **the location**, reporting and **the nature of the event** that has been detected. The number will be redialed every 20 seconds until **telephone connection** is obtained.
- ❖ All **interaction** with SafeHome is managed by **a user interaction subsystem** that reads input provided through the **keypad** and **function keys** ...

Potential Classes for SafeHome

Potential Class	General Classification	Potential Class	General Classification
Homeowner	Role or external entity	Master password	Thing
Sensor	external entity	Telephone number	Thing
Control panel	external entity	Sensor event	occurrence
Installation	occurrence	Audible alarm	External entity
System (security system)	thing	Monitoring service	Organizational Unit or external entity
Number, type	Attributes of a sensor		

Potential Classes

- ❖ ***Retained information.*** The potential class will be useful during analysis only if information about it must be remembered so that the system can function.
- ❖ ***Needed services.*** The potential class must have a set of identifiable operations that can change the value of its attributes in some way.
- ❖ ***Multiple attributes.*** During requirement analysis, the focus should be on "major" information; a class with a single attribute may, in fact, be useful during design, but is probably better represented as an attribute of another class during the analysis activity.
- ❖ ***Common attributes.*** A set of attributes can be defined for the potential class and these attributes apply to all instances of the class.
- ❖ ***Common operations.*** A set of operations can be defined for the potential class and these operations apply to all instances of the class.
- ❖ ***Essential requirements.*** External entities that appear in the problem space and produce or consume information essential to the operation of any solution for the system will almost always be defined as classes in the requirements model.

潜在类判别

- 保留信息。只有当关于它的信息必须记住以使系统能运行时，潜在类才在分析过程中有用。
- 所需服务。潜在类必须有一套可识别的操作，这些操作可以某种方式改变其属性的值。
- 多个属性。在需求分析期间，应关注“主要”信息；具有单个属性的类在设计期间可能有用，但在分析活动期间可能更好地表示为另一个类的属性。
- 公共属性。可以为潜在类定义一组属性，并且这些属性适用于类的所有实例。
- 公共操作。可以为潜在类定义一组操作，并且这些操作适用于类的所有实例。
- 关键需求。在问题空间中出现并生成或消耗对系统运行至关重要的信息的外部实体，几乎总是被定义为需求模型中的类。

Potential Classes for SafeHome

Potential Class		Characteristic number that applies
Homeowner	X	1X, 2 X, 6V
Sensor	V	
Control panel	V	
Installation	X	
System (security system)	V	
Number, type	X	3X
Master password	X	3X
Telephone number	X	3X
Sensor event	V	
Audible alarm	V	
Monitoring service	X	1X, 2 X, 6V

Potential Classes for SafeHome

Potential Class		Characteristic number that applies
Homeowner	V	1v, 2 v, 6V
Sensor	V	
Control panel	V	
Installation	X	
System (security system)	V	
Number, type	X	3X
Master password	X	3X
Telephone number	X	3X
Sensor event	V	
Audible alarm	V	
Monitoring service	X	1X, 2 X, 6V

发现对象类的实例

小王是一个爱书之人，家里各类书籍已过千册，而平时又时常有朋友外借，因此需要一个个人图书管理系统。该系统应该能够将书籍的基本信息按计算机类、非计算机类分别建档，实现按书名、作者、类别、出版社等关键字的组合查询功能。在使用该系统录入新书籍时系统会自动按规则生成书号，可以修改信息，但一经创建就不允许删除。该系统还应该能够对书籍的外借情况进行记录，可对外借情况列表打印。另外，还希望能够对书籍的购买金额、册数按特定时间周期进行统计

筛选备选类

- “小王”、“人”、“家里”很明显是系统外的概念，无须对其建模；
- 而“个人图书管理系统”、“系统”指的就是将要开发的系统，即系统本身，也无须对其进行建模；
- 很明显“书籍”是一个很重要的类，而“书名”、“作者”、“类别”、“出版社”、“书号”则都是用来描述书籍的基本信息的，因此应该作为“书籍”类的属性处理，而“规则”是指书号的生成规则，而书号则是书籍的一个属性，因此“规则”可以作为编写“书籍”类构造函数的指南。
- “基本信息”则是书名、作者、类别等描述书籍的基本信息统称，“关键字”则是代表其中之一，因此无需对其建模；
- “功能”、“新书籍”、“信息”、“记录”都是在描述需求时使用到的一些相关词语，并不是问题域的本质，因此先可以将其淘汰掉；

筛选修选类

- “计算机类”、“非计算机类”是该系统中图书的两大分类，因此应该对其建模，并改名为“计算机类书籍”和“非计算机类书籍”，以减少歧义；
- “外借情况”则是用来表示一次借阅行为，应该成为一个候选类，多个外借情况将组成“外借情况列表”，而外借情况中一个很重要的角色是“朋友”——借阅主体。虽然到本系统中并不需要建立“朋友”的资料库，但考虑到可能会需要列出某个朋友的借阅情况，因此还是将其列为候选类。为了更好地表述，将“外借情况”改名为“借阅记录”，而将“外借情况列表”改名为“借阅记录列表”；
- “购买金额”、“册数”都是统计的结果，都是一个数字，因此不用将其建模，而“特定时限”则是统计的范围，也无需将其建模；不过从这里的分析中，我们可以发现，在该需求描述中隐藏着一个关键类——书籍列表，也就是执行统计的主体

Potential Class		Characteristic number that applies
小王	X	1X, 2 X, 6V
系统	X	
书籍	V	
书名、作者、类别.....	X	3X
计算机类书籍	V	
非计算机类书籍	V	
借阅记录（外借情况）	V	
借阅记录列表（外借情况列表）	V	
购买金额	X	3X
时限	X	5X

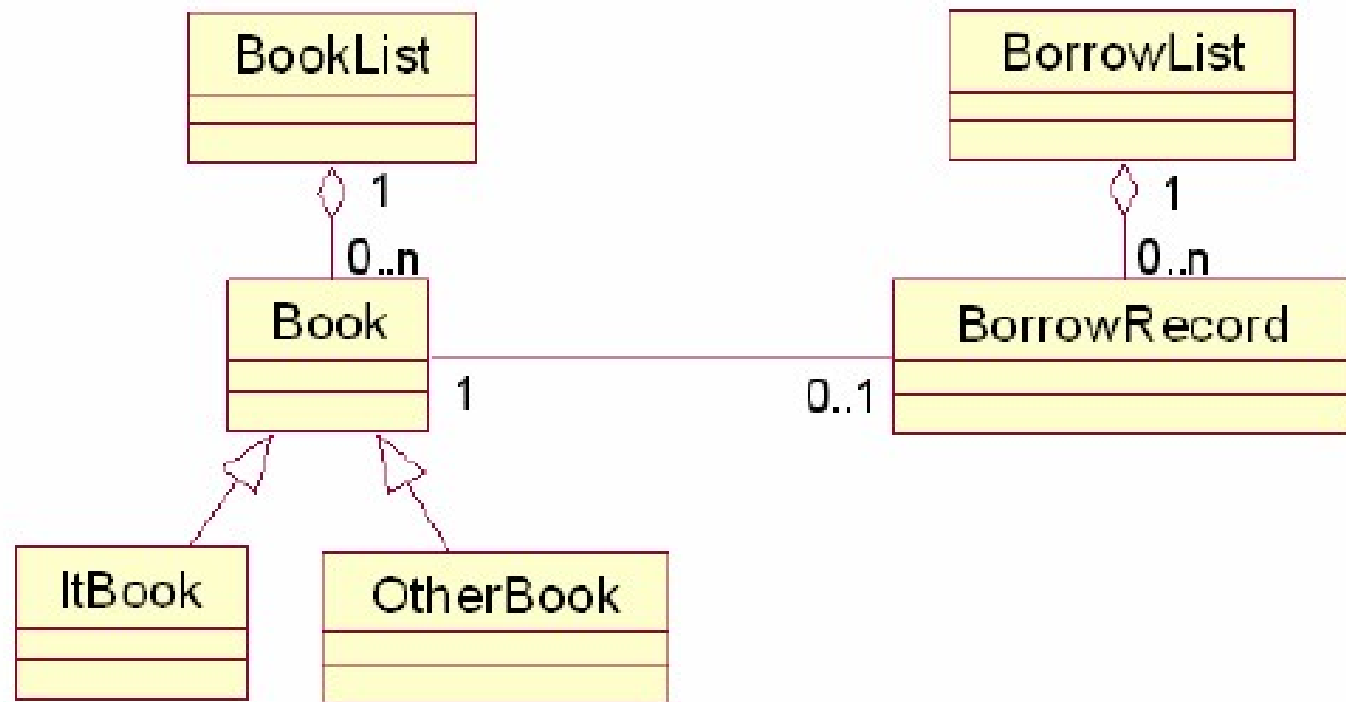
❖ 增补一个“书籍列表”类

得到候选类

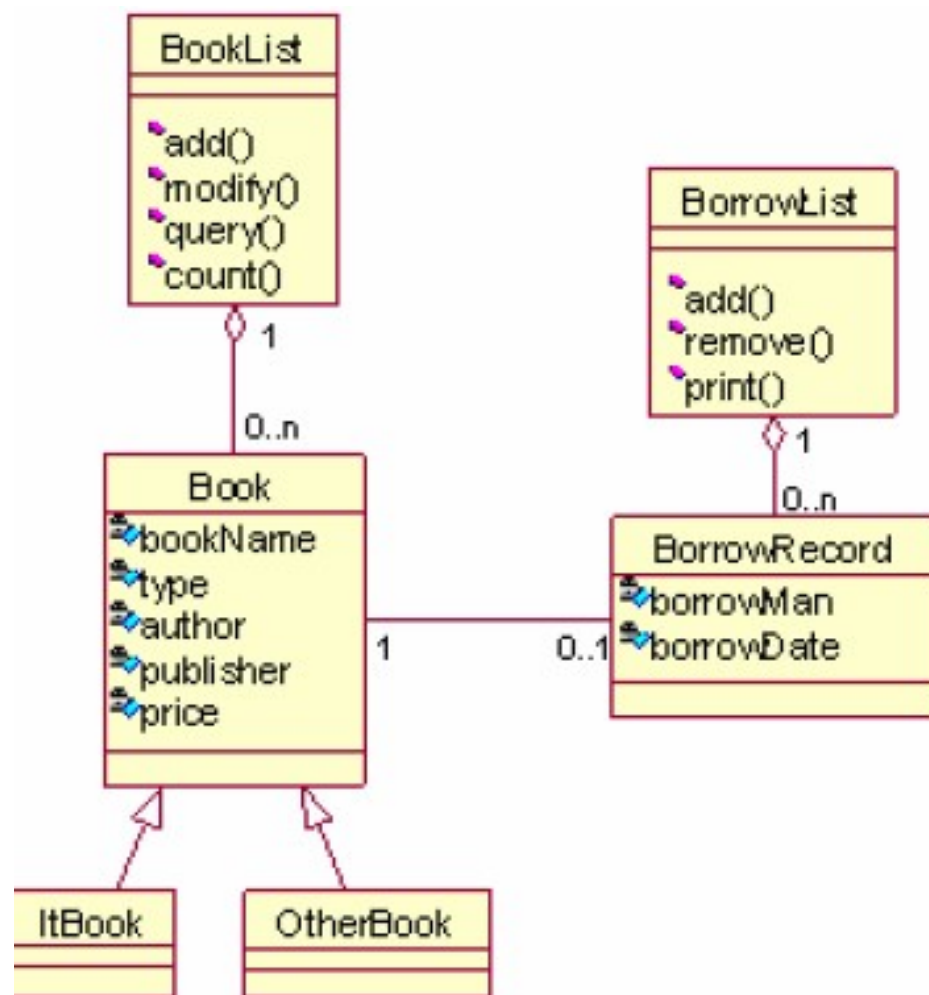
书籍	计算机类书籍	非计算机类书籍
借阅记录	借阅记录列表	书籍列表

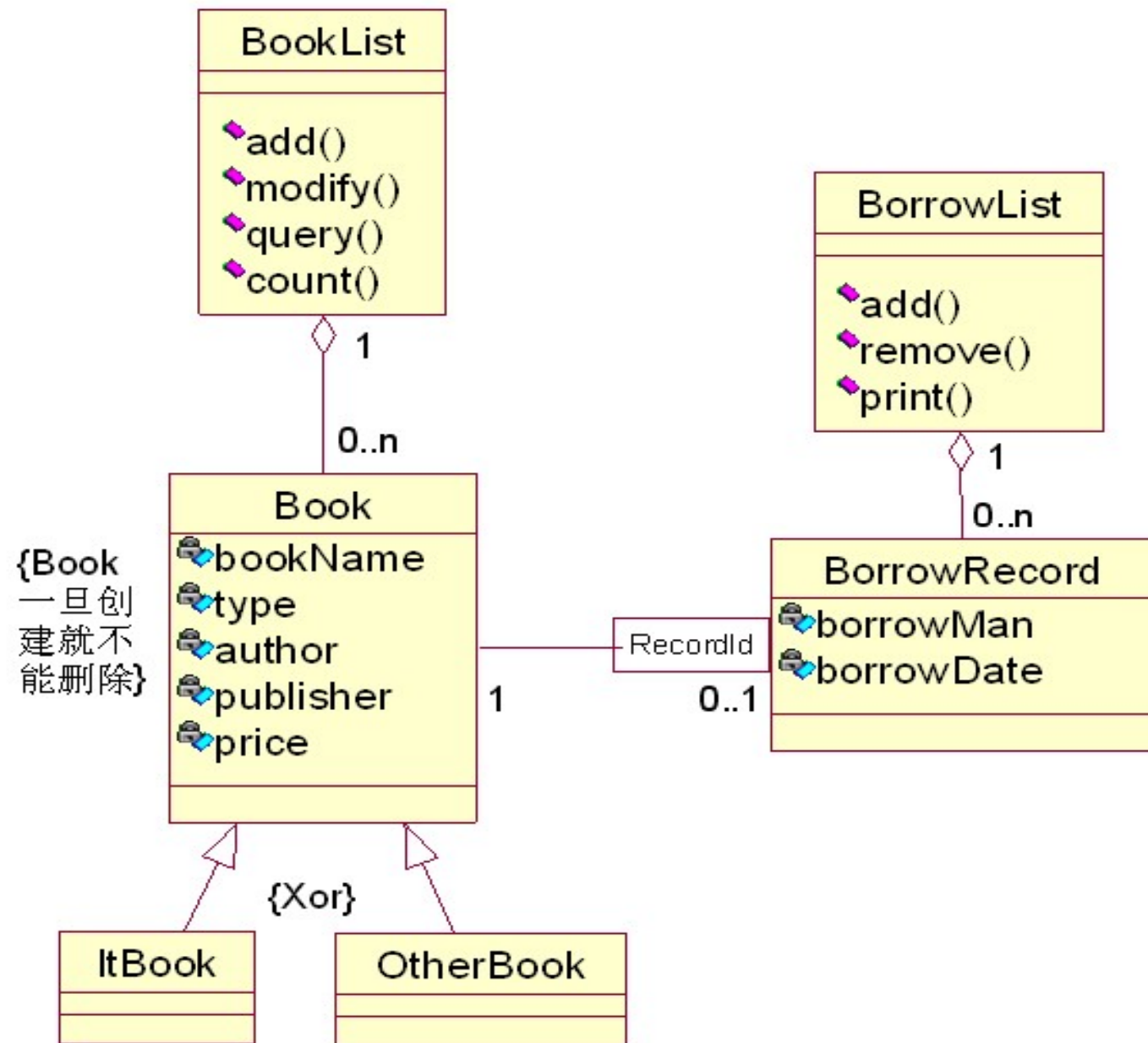
确定对象类

❖ 画出类图



细化对象类





Defining Attributes

❖ *Attributes* describe a class that has been selected for inclusion in the analysis model.

- build two different classes for professional baseball players
 - **For Playing Statistics software:** name, position, batting average, fielding percentage, years played, and games played might be relevant
 - **For Pension Fund (养老金) software:** average salary, credit toward full vesting, pension plan options chosen, mailing address, and the like.
- 属性描述已选定用于包含在分析模型中的类。
- 为职业棒球运动员建立两个不同的类
 - 对于比赛统计软件：姓名、位置、击球率、防守率、比赛年份和比赛场次可能相关
 - 对于养老金（养老金）软件：平均工资、退休金全额归属信用、选择的养老金计划、邮寄地址等

Defining Operations

- ❖ Do a grammatical parse of a processing narrative and look at the verbs
- ❖ Operations can be divided into four broad categories:
 - (1) operations that **manipulate data** in some way (e.g., adding, deleting, reformatting, selecting)
 - (2) operations that perform a **computation**
 - (3) operations that **inquire about the state of an object**, and
 - (4) operations that **monitor an object** for the occurrence of a controlling event.
- 对处理叙述进行语法解析并查看动词
- 操作可以分为四大类：
 - 操作数据某种方式（例如，添加、删除、重新格式化、选择）的操作
 - 执行计算的操作
 - 查询对象状态的操作
 - 监控对象以确定控制事件发生的操作

CRC Models

❖ *Class-responsibility-collaborator (CRC) modeling* [Wir90]

provides a simple means for identifying and organizing the classes that are relevant to system or product requirements. Ambler [Amb95] describes CRC modeling in the following way:

- **A CRC model is really a collection of standard index cards that represent classes.**
 - **The cards are divided into three sections. Along the top of the card you write the name of the class. In the body of the card you list the class responsibilities on the left and the collaborators on the right.**
- 类-责任-协作者（CRC）建模 [Wir90] 提供了一种简单的方法来识别和组织与系统或产品需求相关的类。Ambler [Amb95] 以如下方式描述了CRC建模：
 - CRC模型实际上是一组表示类的标准索引卡片的集合。
 - 卡片分为三部分。在卡片顶部写上类名。在卡片主体中列出类的责任在左边，协作者在右边。

CRC Modeling

ClassFloorPlan	
Description:	
Responsibility:	Collaborator:
defines floor plan name/type	
manages floor plan positioning	
scales floor plan for display	
scales floor plan for display	
incorporates walls, doors and windows	Wall
shows position of video cameras	Camera

Class Types

- 实体类——明确的事物
- 边界类——与用户交互的接口
- 控制类——管理系统的数据走向

- ❖ **Entity classes**, also called **model** or **business classes**, are extracted directly from the statement of the problem (e.g., FloorPlan and Sensor).
- ❖ **Boundary classes** are used to create the interface (e.g., interactive screen or printed reports) that the user sees and interacts with as the software is used.
- ❖ **Controller classes** manage a “unit of work” [UML03] from start to finish. That is, controller classes can be designed to manage
 - the creation or update of entity objects;
 - the instantiation of boundary objects as they obtain information from entity objects;
 - complex communication between sets of objects;
 - validation of data communicated between objects or between the user and the application.

- 实体类，也称为模型或业务类，是直接从问题陈述中提取的（例如，平面图和传感器）
- 边界类用于创建用户在使用软件时看到和交互的界面（例如，交互屏幕或打印报告）。
- 控制类管理“工作单元” [UML03] 从开始到结束。也就是说，控制类可以设计为管理：
 - 实体对象的创建或更新；
 - 获取来自实体对象信息的边界对象的实例化；
 - 对象集之间的复杂通信；
 - 对象之间或用户和应用之间通信的数据验证。

Responsibilities

- 智能系统应该分布在所有类中
- 职责说明应该具有普遍性 (可以继承)
- 信息和操作应该放在同一个类中
- 有关同一事物的应当作为同一个类
- 适当时, 职责可以被共享

- ❖ **System intelligence should be distributed across classes to best address the needs of the problem**
- ❖ **Each responsibility should be stated as generally as possible**
- ❖ **Information and the behavior related to it should reside within the same class**
- ❖ **Information about one thing should be localized with a single class, not distributed across multiple classes.**
- ❖ **Responsibilities should be shared among related classes, when appropriate.**

- 系统智能应分布在各个类中, 以最好地解决问题的需求。
- 每个职责应尽可能笼统地陈述。
- 关于某一事物的信息应集中在一个类中, 而不是分布在多个类中。
- 信息及其相关行为应位于同一类中。
- 相关类之间应在适当时共享职责。

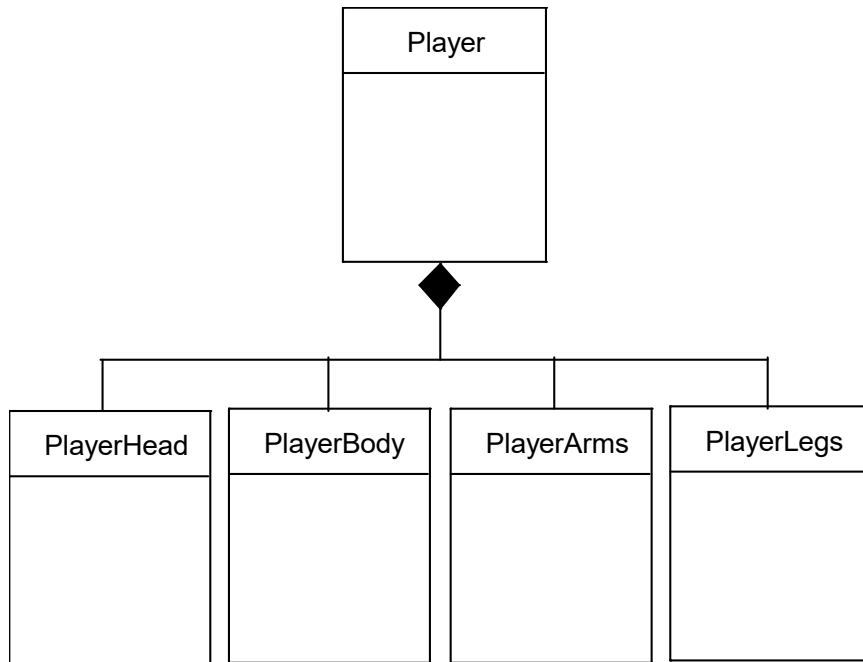
- 类有两种方式实现其职责——1、使用自身的操作控制自身的属性 2、和其他类协作
- 通用的协作关系
 - is-part-of
 - has-knowledge-of
 - depends-upon

Collaborations

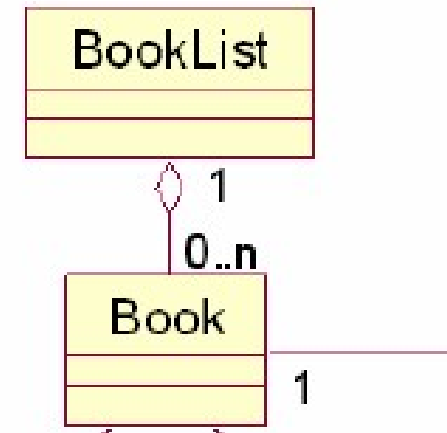
- ❖ **Classes fulfill their responsibilities in one of two ways:**
 - A class can use its own operations to manipulate its own attributes, thereby fulfilling a particular responsibility, or
 - a class can collaborate with other classes.
- ❖ **Collaborations identify relationships between classes**
- ❖ **Collaborations are identified by determining whether a class can fulfill each responsibility itself**
- ❖ **three different generic relationships between classes [WIR90]:**
 - the *is-part-of* relationship
 - the *has-knowledge-of* relationship
 - the *depends-upon* relationship

Composite Aggregate Class

Part of



Has-a



审查CRC模型

Reviewing the CRC Model

- ❖ **All participants in the review (of the CRC model) are given a subset of the CRC model index cards.**
 - Cards that collaborate should be separated (i.e., no reviewer should have two cards that collaborate).
- ❖ **All use-case scenarios (and corresponding use-case diagrams) should be organized into categories.**
- ❖ **The review leader reads the use-case deliberately.**
 - As the review leader comes to a named object, she passes a token to the person holding the corresponding class index card.

-
- ❖ **When the token is passed, the holder of the class card is asked to describe the responsibilities noted on the card.**
 - **The group determines whether one (or more) of the responsibilities satisfies the use-case requirement.**
 - ❖ **If the responsibilities and collaborations noted on the index cards cannot accommodate the use-case, modifications are made to the cards.**
 - **This may include the definition of new classes (and corresponding CRC index cards) or the specification of new or revised responsibilities or collaborations on existing cards.**

审查CRC模型

- 审查中的所有参与者（CRC模型审查）都分到一部分CRC模型索引卡片。
 - 协作的卡片应分开（即，没有审查员应持有两张协作卡片）。
- 所有用例场景（及相应的用例图）应分类组织。
- 审查领导仔细阅读用例。
 - 当审查领导遇到命名对象时，她将一个标记传递给持有相应类索引卡片的人。

当标记传递时，类卡片持有人被要求描述卡片上的职责。

- 小组确定一个（或多个）职责是否满足用例需求。
- 如果索引卡片上的职责和协作无法容纳用例，则对卡片进行修改。
 - 这可能包括定义新类（及相应的CRC索引卡片）或在现有卡片上指定新的或修订的职责或协作。

■ 评审CRC模型

1. 评审员拿到不存在协作关系的CRC卡
2. 分类管理用例场景
3. 评审组长阅读用例，看到已命名的对象后发token给对应的人
4. 令牌传递时被传递者描述卡上的职责，讨论确定是否满足用例
5. 如果不能满足用例则修改卡片

关联与依赖

Associations and Dependencies

❖ Two analysis classes are often related to one another in some fashion

- In UML these relationships are called *associations*
- Associations can be refined by indicating *multiplicity* (the term *cardinality* is used in data modeling)

❖ In many instances, a client-server relationship exists between two analysis classes.

- In such cases, a client-class depends on the server-class in some way and a *dependency relationship* is established

- 两个分析类通常以某种方式相互关联。

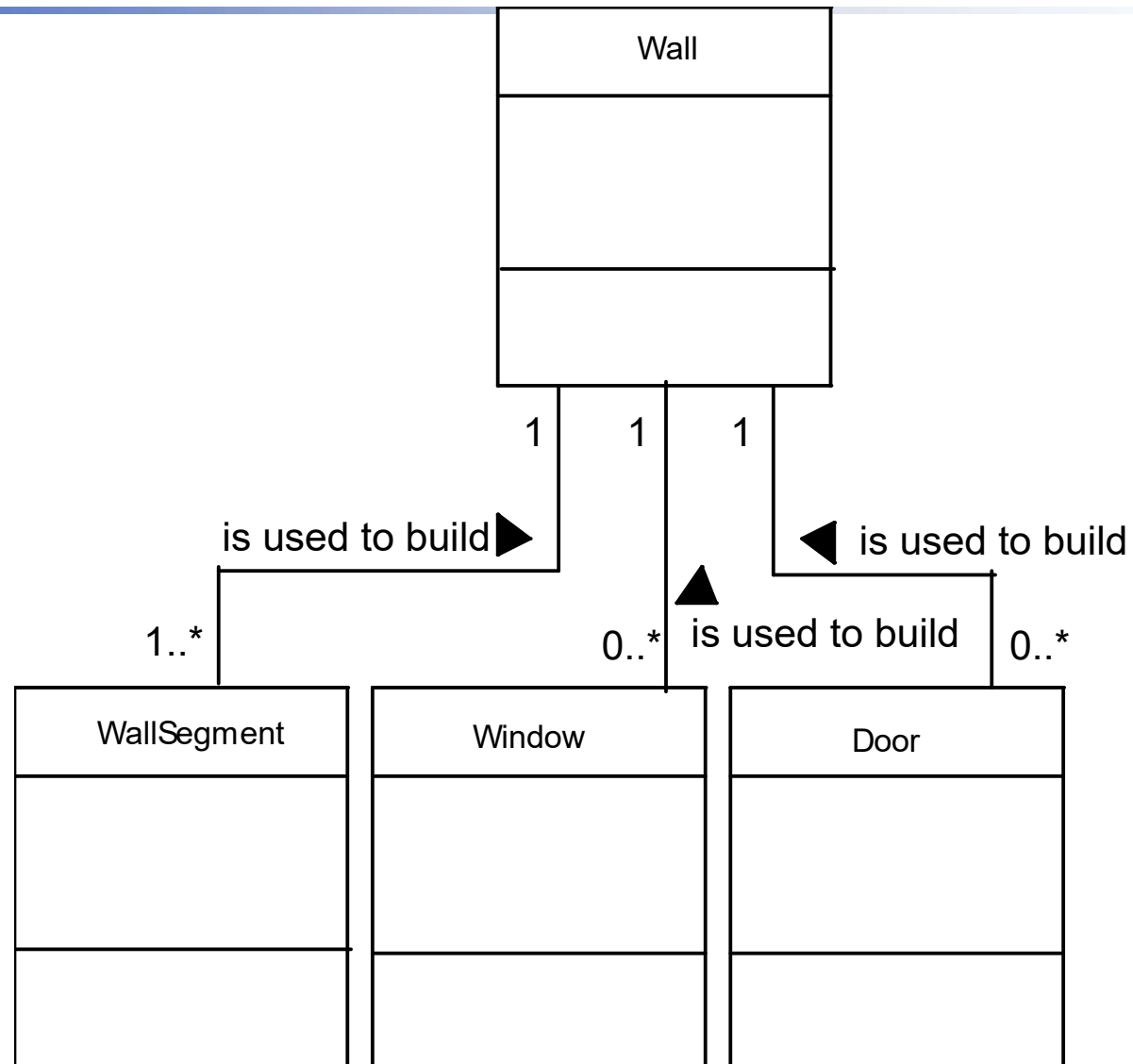
- 在UML中，这些关系称为关联。

- 关联可以通过指示多重性（在数据建模中使用的术语是基数）来细化。

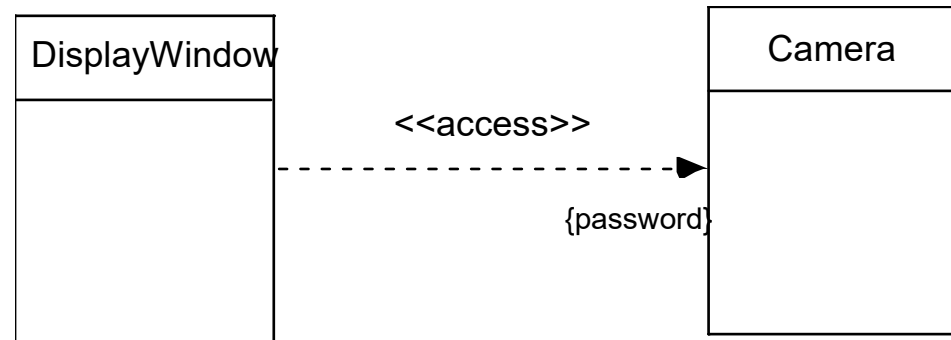
- 在许多情况下，两个分析类之间存在客户端-服务器关系。

- 在这种情况下，客户端类在某种程度上依赖于服务器类，并建立依赖关系。

Multiplicity



Dependencies



Analysis Packages

▪ 分析包——把相同功能的类放到一起

- ❖ Various elements of the analysis model (e.g., use-cases, analysis classes) are categorized in a manner that packages them as a grouping.
- ❖ The plus sign preceding the analysis class name in each package indicates that the classes have public visibility and are therefore accessible from other packages.
- ❖ Other symbols can precede an element within a package. A minus sign indicates that an element is hidden from all other packages and a # symbol indicates that an element is accessible only to packages contained within a given package.

- 分析模型的各种元素（例如，用例、分析类）以将它们分组的方式进行分类。
- 在每个包中的分析类名之前的加号表示这些类具有公共可见性，因此可以从其他包访问。
- 其他符号可以出现在包中的元素之前。减号表示一个元素对所有其他包隐藏，#符号表示一个元素仅对包含在给定包中的包可访问。

Analysis Packages

